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Pipe collar, mold and method

Abstract

Embodiments of the present disclosure provide a pipe collar for a drainage pipe and a mold for the pipe collar, wherein the pipe collar is formed by positioning mold walls adjacent, atop and/or around a mold face, inserting one or more safety reinforcement handlock bar elements through one or more of the walls, filling the mold with a consolidating material and removing the walls. Upon removing the walls, a formed pipe collar can be lifted, moved and otherwise manipulated using the safety reinforcement handlock bar element(s) to position the pipe collar above a drainage pipe.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to pipe collars and, more particularly, to a pipe end wall collar, mold and method for protecting and improving the appearance of drainage pipe ends.

BACKGROUND AND SUMMARY

(2) Drainage pipes and/or culverts are often used in residential settings under a driveway near the street. These drainage pipes form a tunnel permitting rainwater to flow underneath the driveway. Typically, drainage pipes are made of metal or concrete and are embedded in the surrounding soil. Stormwater runoff, drainage buildup and other matter can degrade the qualities and appearance of such drainage pipes. Further, exposed drainage pipes can be bent, crushed or damaged by various types of equipment working around the area where drainage pipes are located. Additionally, unprotected pipe ends can create safety hazards due to jagged, chipped or broken edges of whatever material is employed.

(3) Embodiments of the present disclosure provide a pipe collar, mold and method facilitating the protection of drainage pipes from damage and erosion while providing an aesthetically pleasing appearance. Embodiments of the present disclosure are provided by employing a mold face which may optionally be decorative, assembling mold walls adjacent and/or above the mold face, inserting one or more safety reinforcement handlock bar elements through openings in opposing mold walls and inserting a consolidating material such as concrete atop the mold face and between the mold walls. In various embodiments, the safety reinforcement handlock bar includes one or more sleeves provided around at least one or more portions of rebar, wherein the rebar is maintained within the mold to enhance strength and mobility of the pipe collar. In various embodiments, the mold walls include a substantially semi-circular wall that assists in forming a pipe collar with a semi-circular bottom wall adapted to sit atop and in communication with a drainage pipe.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a mold face and mold walls according to embodiments of the present disclosure.

(2) FIG. 2 is a perspective view of the mold walls positioned around and/or above the mold face in

accordance with embodiments of the present disclosure.

(3) FIG. 3 is a perspective view similar to FIG. 2 that shows openings in opposite walls in accordance with embodiments of the present disclosure.

(4) FIG. 4 is a perspective view similar to FIG. 3 that shows a safety reinforcement handlock bar element extending through the openings in the mold walls in accordance with embodiments of the present disclosure.

(5) FIG. 5 is a perspective view similar to FIG. 4 that shows a consolidating material being poured into the mold in accordance with embodiments of the present disclosure.

(6) FIG. 6 is a perspective view showing the mold filled with the consolidating material in accordance with embodiments of the present disclosure.

(7) FIG. 7 is a perspective view showing directions for removing the mold walls in accordance with embodiments of the present disclosure.

(8) FIG. 8 is a front view of a pipe collar in accordance with embodiments of the present disclosure.

(9) FIGS. 9 and 10 are schematic views showing a drainage pipe both without and with a pipe collar installed in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS

(10) The presently disclosed subject matter now will be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all embodiments of the presently disclosed subject matter are shown. Like numbers refer to like elements throughout. The presently disclosed subject matter may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Indeed, many modifications and other embodiments of the presently disclosed subject matter set forth herein will come to mind to one skilled in the art to which the presently disclosed subject matter pertains having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the presently disclosed subject matter is not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims.

(11) It will be appreciated that reference to “a”, “an” or other indefinite article in the present disclosure encompasses one or a plurality of the described element. Thus, for example, reference to a safety reinforcement handlock bar may encompass one or more safety reinforcement handlock bars, reference to a wall may encompass one or more walls, and so forth.

(12) As shown in FIGS. 1 through 7, a mold 10 is provided for producing a pipe collar as described herein. In various embodiments, the mold 10 includes a mold base 12 having a mold face 15, first 14 and second 16 side edges, a top edge 18 and a bottom edge 20. The mold 10 can also include first 24 and second 26 mold side walls, wherein the first mold side wall 24 is positioned adjacent or atop the first side edge 14 of the mold base 12 and the second mold side wall 26 is positioned adjacent or atop the second side edge 16 of the mold base 12 prior to the pipe collar being formed. The mold 10 can further include a mold bottom wall 30 formed in a rounded or semi-circular shape. The mold bottom wall 30 can be positioned atop the mold face 15 prior to the pipe collar being formed. In various embodiments, the mold face 15 is formed with a physical pattern as indicated by lines 33, and it will be appreciated that the physical pattern 33 on the mold face 15 can cause the inverse pattern to be formed on the front main face of the pipe collar when formed according to the present disclosure, as disclosed elsewhere herein. It will be appreciated that the physical pattern 33, when present, can extend outwardly of the mold face, can be an indentation into the mold face or some combination of an outward extension and an inward indentation, for example.

(13) In various embodiments, the mold 10 further includes a top wall 35 positioned so as to form a gap G (see FIG. 2) between the top wall 35 and the top edge 18 of the mold base 12. This gap G

facilitates the formation of a front top face on the pipe collar when formed, and the front top face can be formed with no pattern, or some pattern that differs from the front main face of the pipe collar when formed, for example. The mold **10** can further include an intermediate wall **37** positioned adjacent the top edge **18** of the mold base **12**. The intermediate wall **37** can function as a guide in various embodiments when the mold walls **24**, **26**, **35** are placed on or around the mold base **12**, as indicated by arrows **55** in FIG. **1**. The mold can also include bottom edge walls **38** positioned between each of the first **24** and second **26** mold side walls and the mold bottom wall **30**.

(14) The mold walls **24**, **26**, **30**, **35**, **37**, **38** can be placed around, atop and/or over the mold base **12** as shown by arrows **55** in FIG. **1**, prior to forming the pipe collar as described herein. FIG. **2** shows an embodiment with the full mold **10** in place, where walls **24**, **26**, **30** and **38** are atop the mold base **12**. Top wall **35** is positioned with a gap **G** between the top wall **35** and the top edge **18** of the mold base **12**. Intermediate wall **37** is positioned behind and adjacent/in contact with the top edge **18** of the mold base **12**. In this way, the intermediate wall **37** aligns the rest of the mold **10** with the mold base **12** for accurate production of the pipe collar as described herein. The mold walls **24**, **26**, **30**, **38**, **35** and/or **37** form a cavity **40** as shown in FIG. **2**.

(15) As shown in FIG. **3**, embodiments of the present disclosure employ openings **42** formed in the first **24** and second **26** mold side walls. These openings **42** are aligned such that a safety reinforcement handlock bar element **44** (see FIGS. **4** and **5**) can be positioned through the openings **42**. The safety reinforcement handlock bar element **44** can be provided in many forms. In various embodiments, the safety reinforcement handlock bar element **44** includes a metal reinforcing bar or rebar **90** having first **91** and second **92** ends, wherein a sleeve **95** is positioned around the first end **91** and a sleeve **96** is positioned around the second end **92**. Each sleeve **95**, **96** is formed to snugly fit within the openings **42** by compression or friction fit. In various embodiments, the rebar **90** does not extend fully through the length **Z** of each sleeve **95**, **96**, but rather only partially through the length **Z**, as shown by the extent of the dashed lines of first end **91** and second end **92** in FIG. **4**. In this way, the rebar **90** can be maintained in stationary position during consolidation of the consolidating material as described hereinafter while also leaving room within the sleeves **95**, **96** for an external element such as a handle, screwdriver or other element (see **48** in FIGS. **7** and **8**) to be inserted into the sleeves **95**, **96** from the outside of the pipe collar **70** to facilitate transfer of the pipe collar **70** once formed, as described elsewhere herein.

(16) In alternative embodiments, the first **91** and second **92** ends of the rebar **90** can extend the length of the sleeves **95**, **96** and/or beyond the length of the sleeves **95**, **96** and outwardly of walls **24**, **26**, wherein the rebar **90** can then be used as a handle for lifting and moving the pipe collar **70**. In still other embodiments, the sleeves **95**, **96** extend outwardly of walls **24**, **26** and the externally extending portion of the sleeves can be used for lifting and moving the pipe collar. In such embodiments, the extending sleeves **95**, **96** and/or extending rebar **90** could be hidden or cut off after installation depending upon desired appearances. In still other embodiments, the safety reinforcement handlock bar **44** is a rebar **90** alone, or a single sleeve alone, wherein the rebar or single sleeve extends from wall **24** to wall **26** and through the openings **42**.

(17) In various embodiments, the openings **42** are positioned a suitable distance below the top edge **45** of each of the first **24** and second **26** mold side walls, as well as a suitable distance inside of the edge **46** of the top wall **35** to enable the safety reinforcement handlock bar **44** to operate as a stabilizing element in the pipe collar once formed as well as to facilitate lifting when the pipe collar **70** is formed and requires manual movement (e.g., onto or off of a loading truck, and/or when being installed or moved).

(18) It will be appreciated that once the mold is assembled and prepared for a consolidating material **50** such as concrete to be poured into the cavity **40**, the intermediate wall **37** shown in FIGS. **1** through **4** can be removed. As shown in FIG. **5**, a suitable dispensing apparatus **52** can pour the consolidating material (e.g., concrete) **50** into the cavity **40** as shown by arrow **54**. With

the intermediate wall **37** removed, the consolidating material can extend to the full range of the cavity **40** and promote a uniformly consistent pipe collar **70** that meets the intended face design **72**, if any, prompted by the physical pattern **33**, if any, on the mold face **15**. The safety reinforcement handlock bar **44** is maintained in a static position after the pipe collar **70** is formed, reinforcing the consolidating material **50** and further providing openings for handles **48** such as screwdrivers or other tools to be inserted in the sleeves **95**, **96** for carrying the pipe collar **70**. With reference to FIG. 7, once the consolidating material has suitably consolidated and dried, the walls **24**, **26**, **30**, **35** and **38** can be removed in various ways that will be understood to those of ordinary skill and the pipe collar **70** can be lifted off of the mold base **12**. It will be appreciated that the handles **48** are inserted after the walls are removed from the formed pipe collar **70**.

(19) The finished pipe collar **70** (with handles **48** inserted) can appear as shown in FIG. 8, for example, and the handles **48** can be employed as described elsewhere herein for moving the pipe collar **70**. Once the pipe collar **70** reaches its intended position, the handles **48** can be removed. The pipe collar **70** is thus formed with a front face **72** and optionally a top face **74**, both of which may or may not have a pattern design. The pipe collar **70** in the embodiment of FIG. 8 is formed with substantially parallel outer side walls **77** due to the positioning of the first **24** and second **26** mold side walls during forming. The pipe collar **70** is further formed with a rounded bottom wall **75** that can substantially mate with a drainage pipe **80**, such as alongside a residential driveway **82** as shown in FIGS. 9 and 10. It will be appreciated that the front top face **74**, if provided with a pattern design, can be a pattern design that is the same or different from the patterned front face **72** of the pipe collar **70**. In various embodiments, the front top face **74** is positioned above the collar patterned face **72** when the pipe collar **70** is in an upright position, as shown in FIG. 8.

(20) In various embodiments, more than one safety reinforcement handlock bar **44** can be provided. In this way, multiple safety reinforcement handlock bars can extend from each side of a finished pipe collar, which can facilitate use of two hands or multiple people on each side assisting with the transport and placement of the pipe collar **70**. In various embodiments, when the front face **72** has a pattern, the patterned front face **72** can comprise a brick pattern, a stone pattern, a wood or timber pattern or a variety of other patterns as desired.

(21) The present disclosure describes numerous embodiments of the present invention, and these embodiments are presented for illustrative purposes only. These embodiments are described in sufficient detail to enable those skilled in the art to practice the invention embodiments, and it will be appreciated that other embodiments may be employed and that structural, logical, software, electrical and other changes may be made without departing from the scope or spirit of the present invention. Accordingly, those skilled in the art will recognize that the present invention may be practiced with various modifications and alterations. Although particular features of the present invention can be described with reference to one or more particular embodiments or figures that form a part of the present disclosure, and in which are shown, by way of illustration, specific embodiments of the invention, it will be appreciated that such features are not limited to usage in the one or more particular embodiments or figures with reference to which they are described. The present disclosure is thus neither a literal description of all embodiments of the invention nor a listing of features of the invention that must be present in all embodiments.

Claims

1. A pipe collar comprising: a concrete collar formed by pouring concrete into a cavity of a mold, wherein the cavity is defined by mold walls positioned around, atop or over a mold base, wherein at least a subset of the mold walls is formed in a rounded or semi-circular shape so as to produce the concrete collar with a rounded or semi-circular bottom wall, wherein the mold base is formed with a mold face, and wherein the mold face faces the concrete collar to form a collar patterned face, wherein the mold walls comprise a top wall positioned so as to form a gap between the top

- wall and the mold base, whereupon the concrete collar comprises a front top face and the collar patterned face; and a safety reinforcement handlock bar extending through the concrete collar.
2. The pipe collar of claim 1, wherein the safety reinforcement handlock bar comprises a rebar positioned within one or more sleeves.
 3. The pipe collar of claim 1, wherein the safety reinforcement handlock bar is maintained in a static position after the concrete collar is formed.
 4. The pipe collar of claim 1, wherein the mold comprises the mold walls and the mold base.
 5. The pipe collar of claim 1, wherein the collar patterned face comprises a brick pattern.
 6. The pipe collar of claim 1, wherein the collar patterned face comprises a stone pattern.
 7. The pipe collar of claim 1, wherein the collar patterned face comprises a timber pattern.
 8. The pipe collar of claim 1, wherein the front top face is positioned above the collar patterned face when the concrete collar is in an upright position.
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